

Effect sizes and association measures

Overview

Effect sizes complement p-values: they answer “how big is the effect?” rather than “is there an effect at all?”. MORIE exposes the standard families used in carceral, public-health, and sociolegal research.

Cohen’s d (continuous, two-group)

```
library(morie)
set.seed(7)
group_a <- rnorm(60, mean = 0.0)
group_b <- rnorm(60, mean = 0.6)

d <- cohens_d(group_a, group_b)
d
#> [1] -0.5043227
```

Cohen’s d expresses the difference between two means in pooled standard-deviation units. Conventional benchmarks: 0.2 small, 0.5 medium, 0.8 large.

Cramer’s V (categorical association)

```
tab <- matrix(c(20, 10, 30,
                15, 25, 35), nrow = 2, byrow = TRUE)
v <- cramers_v(tab)
v
#> [1] 0.209657
```

Cramer’s V scales the chi-square statistic to a 0–1 association measure for contingency tables. It is particularly useful for the provincial-vs-federal Mandela-rate cross-comparisons in MRM.

Omega-squared (one-way ANOVA)

Omega-squared is a less-biased effect-size estimator than eta-squared for a one-way ANOVA design.

```
omega_squared(f_stat = 5.2, df_between = 2, df_within = 87, n = 90)
#> [1] 0.08536585
```

Proportion confidence intervals

For a binomial proportion, MORIE exposes Wilson, Clopper–Pearson exact, and Wald CIs. Wilson is the default and is what we recommend in published papers (it has better small-sample coverage than Wald).

```
proportion_ci(35, 100)                                # Wilson, 95% CI
#> $p_hat
#> [1] 0.35
#>
#> $ci_lower
#> [1] 0.2636425
#>
#> $ci_upper
#> [1] 0.4474556
proportion_ci(35, 100, method = "exact")              # Clopper-Pearson
#> $p_hat
#> [1] 0.35
#>
#> $ci_lower
#> [1] 0.2572938
#>
#> $ci_upper
#> [1] 0.4518494
proportion_ci(35, 100, method = "wald")              # Wald
#> $p_hat
#> [1] 0.35
#>
#> $ci_lower
#> [1] 0.2565157
#>
#> $ci_upper
#> [1] 0.4434843
```

E-value (sensitivity analysis)

```
e_value(rr = 2.0)
#> $e_value
#> [1] 3.414214
#>
#> $e_value_ci
#> [1] NA
```

The E-value is the minimum strength of association that an unmeasured confounder would need on both treatment and outcome to fully explain away an observed risk ratio.

Where to go next

- For survey-weighted versions of `cohens_d` and `cramers_v`, see the `survey-weighted` vignette.

- For full causal-inference workflows that include ATE / ATT / ATC effect sizes with CIs, see the `causal-inference` vignette.